Tapered kaleidoscope — 100 mm prototype

Cutting and assembly

Three identical printed rails, each carrying one flat trapezoidal mirror, lapped together at the corners into a truncated triangular pyramid. Because the three mirror planes converge on a common apex 100 mm beyond the narrow end, the images tile a *sphere* rather than the flat honeycomb an untapered scope produces. You look in the wide end; the glass chips go at the narrow end.

Mirror-face triangle	25.40 mm at the narrow end, 50.80 mm at the wide end
Length along the axis	100 mm
Taper half-angle	4.1936°
Corner dihedral	60.529286° *(not 60 — the excess is the whole effect)*
Image	a sphere subtending 60°, about 453 facets
Optical face	25.40 mm at the narrow end, 50.80 mm at the wide end
Mirror blank	trapezoid, parallel edges 26.40 and 51.80, height 100.27, legs 101.07 mm (the face plus 1.00 mm of overlap allowance — see §4)

1. Print the rails

Three copies of the same part — there is no left or right hand. Laid flat the part is **60.1 × 100.3 mm and 6.54 mm tall**, so all three fit on an MK3 bed at once (190.3 × 100.3 mm), and printing them together keeps all three under identical conditions.

The model already sits with its outside face on the XY plane. Its steepest overhang is 29.5° — the underside of the tongue — against a 55° threshold, and the only near-horizontal face is the groove ceiling, which is a 0.95 mm bridge rather than an overhang. **No supports, and none of this is marginal.**

PrusaSlicer, MK3 with PLA, 0.4 mm nozzle. Start from *0.15mm QUALITY @MK3* and change only these:

Setting	Value	Why
Ironing	On, *topmost surface only*	the pocket floor is the mirror seat
Perimeters	3	makes the 3 mm wall nearly solid
Elephant foot compensation	0.2 mm (verify)	split faces run down to the bed
Perimeter generator	Arachne (default)	handles the 1.2 mm tongue
Layer height	0.15 mm	44 layers; the part is all flat planes
Fill density	15 %	fine as-is; perimeters do the work
Supports	None	29.5° maximum overhang
Brim	None	60 × 100 mm flat footprint, no warp risk
Seam position	Rear or Aligned	cosmetic only, on the outside face

Only the first three are worth arguing about.

Ironing matters most. The pocket floor is a *top* surface, and at 1.59 mm your mirror is thin enough to conform to whatever that floor does — print texture becomes optical texture. Ironing is what stops that. With rigid mirror stock it would not matter.

Three perimeters rather than two. At 0.45 mm extrusion width that makes the 3 mm wall effectively solid. On a part this thin, perimeters do far more for stiffness and for tongue-and-groove strength than infill does — which is also why 15 % fill needs no change.

Elephant foot compensation is not cosmetic here. The split face and both mating faces run right down to the bed, so first-layer squish-out directly prevents the corners closing. 0.2 mm is the Prusa default; just confirm it has not been zeroed.

Dropping to 0.20 mm layers (33 layers) is fine if you would rather iterate faster — the geometry is entirely flat planes and ironing works independently of layer height. STL export quality is irrelevant for the same reason: the part has no curved surfaces at all, so even the coarsest tessellation is exact.

Print a joint test coupon first. All three rails are identical, so if the tongue-and-groove clearance is wrong, all three are wrong together — and at 0.15 mm clearance with a 0.4 mm nozzle that fit is the likeliest thing to need adjusting, since a 1.2 mm tongue is under three extrusions wide and the usual bias is tongues fat and grooves narrow. Use PrusaSlicer's *Cut* tool to take a 20 mm slice off the model, keep one half, add a second instance, print the pair. Ten minutes instead of two hours, and you learn whether to open `tongueClearance` to 0.2 first.

Do not shortcut this by regenerating a 20 mm part in Onshape — that changes the taper completely and gives you a different triangle. Cut the 100 mm model.

2. About the mirror stock

The cheap "unbreakable" tiles are plastic, not glass, so ignore anything you read about glass cutters and running pliers. Two honest warnings:

- **They are sold for safety, not optics, and are usually noticeably wavy.** Flatness is the single most important optical property of a kaleidoscope mirror, so expect a soft, smeary image with visible distortion in the outer facets. Do not judge the design by what these produce — judge the fit, the corners and the workflow.
- **The panels are 1/16″**, so `mirrorThickness` is 1.5875 mm and the pocket comes out at 1.7875 mm. That leaves 0.2 mm of headroom, and the error is one-sided: stock thinner than nominal is harmless, because the bond layer takes up the slack. Only stock more than 0.2 mm *over* nominal causes trouble, by holding the mirror faces proud so the corners interfere.

3. Cutting the mirrors

A jeweller's saw. Because the one optically critical edge comes off the tile already made, none of the sawn edges needs to be precise — they only need to be straight enough to sit inside a ±1 mm window and free of burrs. A saw suits that: it works right up to the factory edge, puts no stress on the sheet, and thin plastic gives it no trouble.

Blade: a #2 to start. The governing rule is two to three teeth engaged in the material at all times, which at 1.5875 mm puts the usable window at #1 through #4:

Blade	≈TPI	Teeth engaged	
#1	51	3.2	smoother edge, closer to clogging
#2	**43**	**2.7**	**start here**
#4	38	2.4	go here if it gums up

Nothing finer than #1/0 — the reflex reach for 4/0 or 6/0 comes from sawing metal, and at this thickness those pack solid with melted plastic almost at once. Nothing coarser than #4 either: below two teeth engaged the blade grabs on the downstroke and can crack a thin sheet. Skip-tooth blades,

sometimes sold as wax blades, are better again for plastic — every second tooth removed, so twice the chip clearance. Buy a dozen; they are pennies each and you will snap some.

1. **Check the template.** Print `mirror-template.pdf` at 100 % — *Actual size*, not *Fit to page* — and verify both rulers on it with a real ruler. If they are off, every dimension is off.

2. **Put the leading edge on a factory edge of the tile.** This is the single biggest thing you can do for image quality, and it costs nothing. Only one edge of each mirror is optically critical — the leading edge, which lands on its neighbour at the seam — and its coating has to run right to the edge. No cut you make will beat the edge the tile was made with. The template is laid out for this: its heavy red line goes on the factory edge, and you cut only the other three sides. Your tiles are 11.75″ (298.45 mm) square and each blank needs 104.4 mm of edge, so two go on one edge and the third on another. There are four edges per tile and four tiles.

3. **Coated side up.** These are rear-surface mirrors: clear plastic in front, the reflective coating on the back. A jeweller's saw cuts on the downstroke, so the teeth enter the top face and break out of the bottom — coated side up puts that breakout on the clear face. It matters less here, since no sawn edge is critical, but one rule for everything beats reasoning it out each time.

4. **Leave the protective film on** and mark on it.

5. **Transfer the marks.** Lay the template so its heavy red line sits on a factory edge, with the strip below the line overhanging so you can sight the alignment, and tape it down. Prick through the centre of all eight circled marks with a pin, lift the paper, and join the pricks with the straightedge. Then move the template to another factory edge and do the third blank the same way.

6. **Fit the blade teeth-down.** They point toward the handle; run a fingernail along the blade and it should catch on the upstroke. Tension it until it pings when plucked — a slack blade wanders and snaps.

7. **Support the work within a few millimetres of the cut.** This is the one that breaks blanks. Unsupported, 1.6 mm sheet flexes on every stroke, the blade grabs, and the piece cracks. A bench pin is ideal; otherwise clamp the tile between two pieces of scrap hardboard right up against the cut line.

8. **Saw the two short ends first,** each starting at the factory edge and running 26.4 and 51.8 mm inward, then the long overlap edge last with the *blank* inside the saw frame's throat rather than the tile body. The blank never sits more than 51.4 mm from that cut line, so a 3″ frame clears it comfortably.

9. **Long slow strokes, light pressure.** Use the whole blade so heat spreads along it; speed is what melts plastic, not pressure. Keep the blade vertical and do not twist it. Cut on the waste side of the line — the kerf is about 0.3 mm and the width window absorbs it without thinking.

10. **Deburr the cut edges only. 400-grit silicon carbide** (wet-or-dry paper — the dark grey kind, not the tan garnet paper, which loads up on plastic), laid flat on a hard surface. Hold the blank vertical, two or three light passes per edge, then **600-grit** if you want it smooth. Keep the edge square — do not round it over, and do not sand onto the reflective face. Dry is fine at this scale; a drop of water stops it clogging if you keep going.

11. **Never sand the factory edge.** Sanding it is exactly the damage the factory edge was chosen to avoid — it rounds the edge and tears the coating back. It needs nothing.

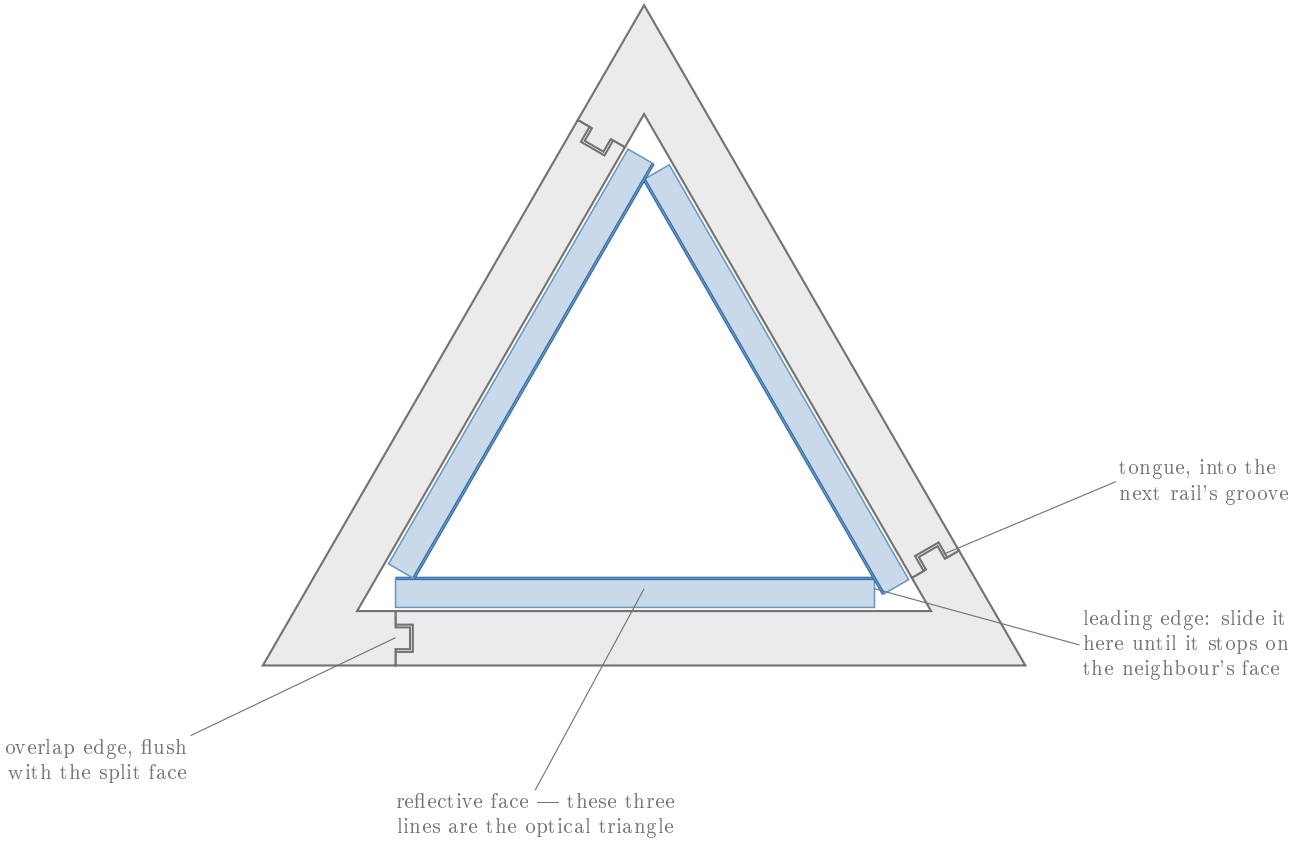
12. **Mark the factory edge** on the coated back as soon as the blank is free, with a stripe of marker. Once all four edges are cut apart it is easy to lose track of which one it was, and at assembly it has to end up on the corner-block side of its rail.

13. **Check the coating.** Hold each slanted edge up to the light. The silver must run right to the edge; any dead band there prints as a dark web across every facet. The factory edge should be perfect — if it is not, the tile was damaged in shipping and you should use a different edge.

Only one of the four edges is optically critical. The leading edge — the factory one — lands on its neighbour at the seam and must carry coating right to its edge. The overlap edge is buried in dead space behind the next mirror and never sees light, so it needs only to be free of burrs. The 26.40 and 51.80 mm ends hide under the end caps and can be rough. If a blank drifts a few tenths wide, leave it; the overlap allowance absorbs it (§4).

4. Where the mirror sits

narrow-end cross-section, 2.4:1



Beyond each ideal vertex there is a **120° void** reaching out to the floor corner, occupied by neither mirror nor rail. The floor runs continuously across the joint there — the neighbouring rail's corner block provides the far half — so that void will carry a mirror extended past the vertex. The blanks are cut 1.00 mm oversize to use it.

Each mirror therefore has two different edges:

- its **leading edge**, on the corner-block side, which stops against the neighbouring mirror's face exactly at the ideal vertex;

- its **overlap edge**, on the split-face side, whose surplus buries itself in the void behind the mirror ahead of it.

The direction is not arbitrary. At each corner it must be the mirror whose face *rises* from the vertex that carries the overlap; then the other mirror's square edge slides in and touches with its top corner first, landing precisely on the vertex. A positive stop. Reverse it and the two simply collide.

The clear face goes inward. On a rear-surface mirror the reflection happens at the coating, but the surface the model positions is the front of the plastic — so the clear face points into the light path and the coated back sits on the bond layer against the pocket floor. Any scratch on that clear face scatters light directly.

The rule: slide each mirror toward its corner-block side until it stops. If the blank is cut to size its overlap edge then finishes flush with the split face, which is a free visual check — and that flush relationship holds no matter what mirror thickness you use, because the split face sits at exactly (face width)/2 plus `edgeMargin`.

Anything from 25.45 to 27.46 mm across the narrow edge will seat. Err generous and sand back: too wide is harmless right up to 2.06 mm of overlap, where the surplus runs out of void, while too narrow eventually leaves the leading edge short of its vertex — and that gap prints as a dark line across every facet.

Orientation of the blank does not matter. Each trapezoid is symmetric, so either slanted edge can serve as the overlap. What must be consistent is that all three mirrors get pushed the same way round.

5. Dry-assemble before you glue anything

Mirrors sitting loose in the pockets, three rails pressed together, masking tape around the outside. Look through it. Thirty seconds of feedback on whether the corners close and the pattern coheres, with nothing committed.

What to look for:

- **Do the tongues seat in the grooves** and the three rails close into a triangle without a gap at the last joint?
- **Does every leading edge actually reach its vertex?** Slide each mirror toward its corner-block side until it stops. A dark line at every cell boundary means a leading edge is short, or the coating has pulled back from the edge.
- **Is there a sphere?** It should read as a ball floating at the far end, about 60° across, not as a flat honeycomb. If it looks flat, something is wrong with the taper, not with your cutting.

6. Bond and close up

Use a neutral-cure silicone. The label must say *neutral cure*, *low odour*, or *mirror safe*; purpose-made mirror mastic also works. Clear rather than white, so squeeze-out does not show.

Do not use ordinary bathroom or plumbing silicone. Most of it is acetoxy-cure — it smells strongly of vinegar as it sets, because it releases acetic acid. Sealed inside a closed mirror tube that acid has nowhere to go, and it attacks the reflective coating at exactly the place you cannot afford to lose it: the freshly cut edges, where no paint layer protects it. The seams darken over weeks. If you are unsure what a tube is, open it and smell: vinegar means acetoxy, so keep it for the bathroom.

Do not use cyanoacrylate. Superglue blooms — it off-gasses and redeposits as a white haze on every nearby surface, which here means the inside of all three mirrors. It also grabs instantly, and this assembly needs the mirrors to stay movable.

Tape is out for the same reason: it sets the position the moment it touches, and the mirrors have to be nudged home after the rails are together. Silicone stays workable long enough.

The model assumes a 0.2 mm bond layer and silicone squeezes down to whatever the pocket leaves. A uniform bond layer only resizes the image sphere very slightly, but an uneven one tilts a mirror, which is the error that actually matters — so run a thin continuous bead down the centre of each pocket rather than dots. A thin mirror bonded on dots will dish between them.

Silicone grips PLA only moderately well, especially against an ironed surface. For a prototype that is a feature: the mirrors peel off and the rails are reusable.

Sequence:

1. A thin bead of silicone along each rail's pocket floor, mirrors laid on roughly in position, all three the same way round.
2. Assemble the three rails, tongue into groove.
3. Reach in through either open end with a thin stick and push each mirror toward its corner-block side until it stops against its neighbour. It will stop — that contact is a positive one, not a judgement call.
4. Band or tape the assembly and leave it to cure.

Then glue the three corner joints and add the object chamber and eyepiece.

Companion files: `mirror-template.pdf` (1:1 cutting template), `kaleidoscope-side.fs` (the Onshape feature, with the geometry derivations in its header), `scope-sim.html` (ray-traced preview — enter dihedral 60.5293 and ratio 2 to see what this prototype should look like).